

What is Pandas

Pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language.

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Pandas Series

A Pandas Series is like a column in a table. It is a 1-D array holding data of any type.

Importing Pandas

```
In [4]: import numpy as np
import pandas as pd
```

Series using String

```
In [6]: # string
country = ['India', 'Pakistan', 'USA', 'Nepal', 'Srilanka']
pd.Series(country)
```

```
Out[6]: 0      India
1    Pakistan
2         USA
3       Nepal
4     Srilanka
dtype: object
```

```
In [7]: # integers
marks= [13,24,56,78,100]
pd.Series(marks)
```

```
Out[7]: 0      13
        1      24
        2      56
        3      78
        4     100
        dtype: int64
```

```
In [8]: # custom index
marks = [67,57,89,100]
subjects = ['maths','english','science','hindi']

pd.Series(marks,index=subjects)
```

```
Out[8]: maths      67
        english    57
        science    89
        hindi     100
        dtype: int64
```

```
In [10]: # setting a name
marks = pd.Series(marks , index=subjects , name="Jack Marks")
marks
```

```
Out[10]: maths      67
         english    57
         science    89
         hindi     100
         Name: Jack Marks, dtype: int64
```

Series from dictionary

When a Pandas Series is converted to a dictionary using the `to_dict()` method, the resulting dictionary has the same keys and values as the Series. The index values of the Series become the keys in the dictionary, and the corresponding values become the values in the dictionary.

```
In [11]: marks = {
          'maths':67,
          'english':57,
          'science':89,
          'hindi':100
        }
marks_series = pd.Series(marks,name="jack Marks")
```

```
In [12]: marks_series
```

```
Out[12]: maths      67  
english    57  
science    89  
hindi     100  
Name: jack Marks, dtype: int64
```

Series Attributes

size: Returns the number of elements in the Series.

```
In [13]: # size  
marks_series.size
```

```
Out[13]: 4
```

dtype: Returns the data type of the values in the Series.

```
In [14]: # dtype  
marks_series.dtype
```

```
Out[14]: dtype('int64')
```

name: Returns the name of the Series.

```
In [15]: # name  
marks_series.name
```

```
Out[15]: 'jack Marks'
```

unique is an attribute of a Pandas Series that returns an array of the unique values in the Series.

```
In [16]: # is_unique  
marks_series.is_unique
```

```
Out[16]: True
```

```
In [17]: pd.Series([1,1,2,3,4,44,2]).is_unique #It gives false because of repetation
```

```
Out[17]: False
```

index: Returns the index labels of the Series.